B.TECH. COMPUTER SCIENCE AND ENGINEERING 2024

PROBLEM SOLVING TECHNIQUES

CourseCode:24CSEI01 **Credits:**02
L:T:P:E0:0:4:0 **ContactHours:**60
Prerequisite(s): NIL

CourseObjectives:
The objective of this course is to

- Work with an IDE to create, edit, compile, run and debug programs
- Analyze the various steps in program development.
- Develop programs to solve basic problems by understanding basic concepts in C like operators, control statements etc.
- Develop modular, reuable and readable C Programs using the concepts like functions, arrays etc.
- Create, read from and write to text and binary files

Course Outcomes
At the end of the course, students will be able to

Course Outcomes	Description	Bloom'sTaxonomy Level
CO1	Apply the specification of syntax rules for numerical constants and variables, datatypes.	Applying(3)
CO2	Implement C programs using decision making, branching, Looping constructs.	Applying(3)
CO3	Develop C Programs using String, Pointers, Files, Structure.	Applying(3)
CO4	Develop C programs to implement one dimensional and two dimensional arrays	Applying(3)
CO5	Implement real time applications using the power of C language features.	Applying(3)
CO6	Analyze the various steps in program development.	Analyzing(4)

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Course Contents

Module-1 **[12 hours]**

PROGRAMMING LANGUAGES AND FUNDAMENTALS OF C

Introduction – Fundamentals of digital computers - Problem Solving Techniques: Algorithm – Flow Chart - Pseudo code, Introduction to Programming – Introduction to C - Structure of Cprogram – Compilation andExecution - CTokens – Keywords, Variables, Constants - DataTypes–Input/outputstatements.

Module-2 **[12hours]**

OPERATORS, DECISION MAKING AND LOOPING STATEMENTS

Operators: Precedence and Associativity-Expressions–Type Conversion and Type casting-Decision making and Branching -Looping-Break and continuestatements.

Module-3 **[12hours]**

ARRAYS AND STRINGS

Introduction to Arrays: Declaration, Initialization – One dimensional array –Two dimensional arrays–Matrix Operations(Addition, Multiplication and Transpose)–Strings-String operations:length, compare, concatenate, copy, reverse – Array of Strings.

Module-4 **[12hours]**

FUNCTION AND POINTERS

Introduction to functions – User – defined functions – Function prototype, function definition, functioncall – Parameter passing: Pass by value, Pass by reference - Recursion. Pointers – Declaration – Pointer expression and Pointer arithmetic – Arrays and pointers – Array of pointers –Function Pointers.

Module-5 **[12hours]**

STRUCTURES AND FILES

Structure – Nested structures –Pointer and Structures – Array of structures – Structure and Functions – Introduction to Files – Using Files in C – Reading and writing Files – Functions for selecting are cord-Commandline arguments-Storage classes– Pre-processor directives-Dynamic memory allocation.

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LabCourseContents:

Lab Exp#	Details
1	a) Write a simple program that converts one given data type to another using auto conversion and casting.Take the values form standard input. b) Write a program for finding the max and min from the three numbers. c) Write the program for the simple,compound interest. d) Write a program that declares Class awarded for a given percentage of marks, where mark <40%= Failed, 40% to <60% = Second class, 60% to <70%=Firstclass,>=70%=Distinction. Readpercentagefromstandardinput. e) Write a program that prints a multiplication table for a given number and the number of rows in the table. For example, for a number 5 and rows = 3, the output should be:e.5 x1 =5 f.5x2 =10g. 5 x3 =15 h. f) Write a program that shows the binary equivalent of a given positive number between 0 to255. g) Write a C program to find sum and average of three numbers.
2	a) Write a C program, which takes two integer roper and sand one operator from the user, performs the operation and then prints the result.(Consider the operators +,-,*,/,% and use Switch Statement). b) Write a C program to find the sum of individual digits of a positive integer and test given number is palindrome. c) A Fibonacci sequence is defined as follows: the first an second terms in thesequence are 0 and 1. Subsequent terms are found by adding the preceding twoterms in the sequence. Write a C program to generate the first n terms of thesequence. d) Write a C program to generate all the prime numbers between 1and n,where n is a value supplied by theuser.

	e) Write a C program to find the roots of a Quadratic equation. f) Write a C program to find the sum of individual digits of a given positive integer. g) Write a C program to check whether given number is Armstrong Number or not. h) Write a C program to check whether given number is perfect number or not. i) Write a C program to check whether given number is strong number or not.
3	a) Write a C program to find the minimum, maximum and average in an array of integers. b) Write a functions to compute mean, variance, Standard Deviation, sorting of n elements in single dimension array. c) Write a Cprogram that uses functions to perform the following: i. Addition of Two Matrices ii. Multiplication of TwoMatrices iii. Transpose of a matrix with memory dynamicall yallocated for then ew matrix as row and column counts may not be same

4	a) Write a C program that converts a number ranging from 1 to 50 to Romanequivalent b) Write a C program that uses functions to perform the following operations: i. To insert a sub-string into a given main string from a given position. ii. To delete n Characters from a given position in a given string. c) Write a C program to determine if thegiven stringis a palindrome or not(Spelled same in both directions with or without a meaning like madam, civic,noon,abcba,etc.) d) Write a C program that displays the position of a character ch in the string S or –1 ifS doesn‘t contain ch. e) Write a C program to count the lines,words and characters in a given text.
5	d) Write C programs that use both recursive and non-recursive functions. i. To find the factorial of a given integer.i. ii. TofindtheGCD(greatestcommondivisor)oftwogivenintegers.j. iii. Tofindx^n
6	a) Write a program for reading elements using pointer into array and display the values using array. b) Write a program for display values reverse order from array using pointer. c) Write a program through pointer variable to sum of n elements from array.

7	a) Write a C program to Calculate Total and Percentage marks of a student usingstructure b) Write a C program to display the contents of a file to standard output device. c) Write a C program which copies one file to another, replacing all lowercase characters with the isuppercase equivalents. d) Write a C program to count thenumber of timesa character occurs in a textfile.The filename and the character are supplied as command line arguments. e) Write a C program that does the following: First, create a binary file and store 10 integers, where the filename and 10 values are given in the command line. (hint: convert the strings using a toi function) Now the program asks for an index and a value from the user, and the value at that index should be changed to the new value in the file. (hint: use fseek function) The program should then read all 10 values and print them back. f) Write a C program to merge two files into a third file (i.e., the contents of the first file followed by those of the second file into a third file).

TextBooks:

1) HerbertSchildt,C:TheCompleteReference,McGrawHill,4thEdition,2017.
2) ProgramminginC,StephenG.Kochan,FourthEdition,PearsonEducation,2015.
3) B.A.ForouzanandR.F.GilbergCProgrammingandDataLearni Structures, Cengage ng,(3rdEdition),2010.